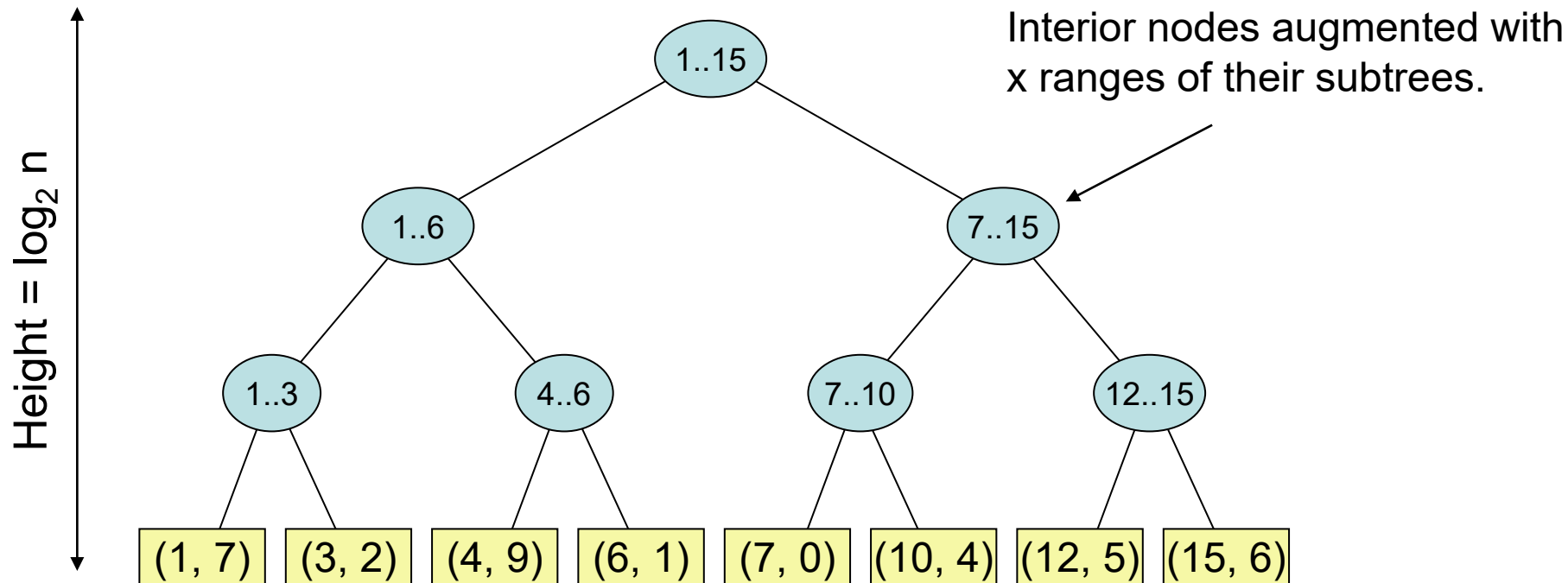


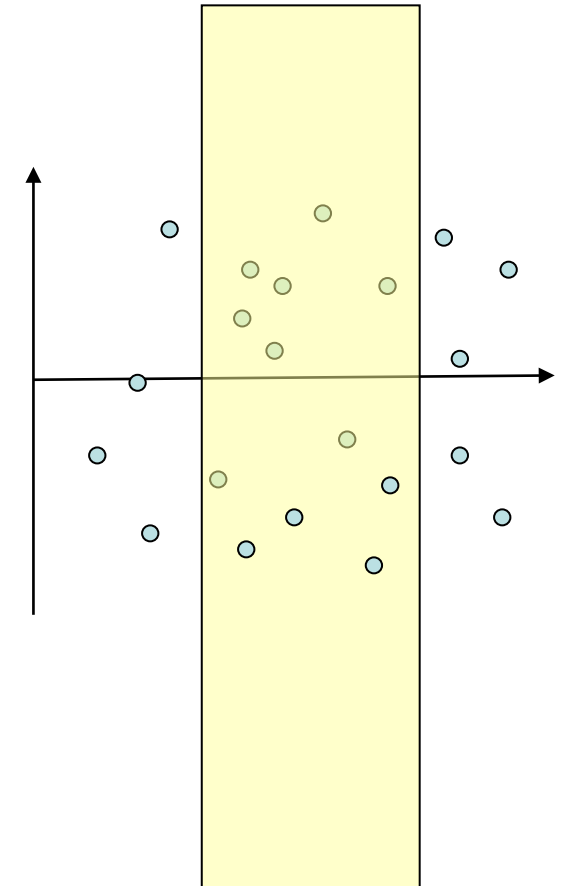
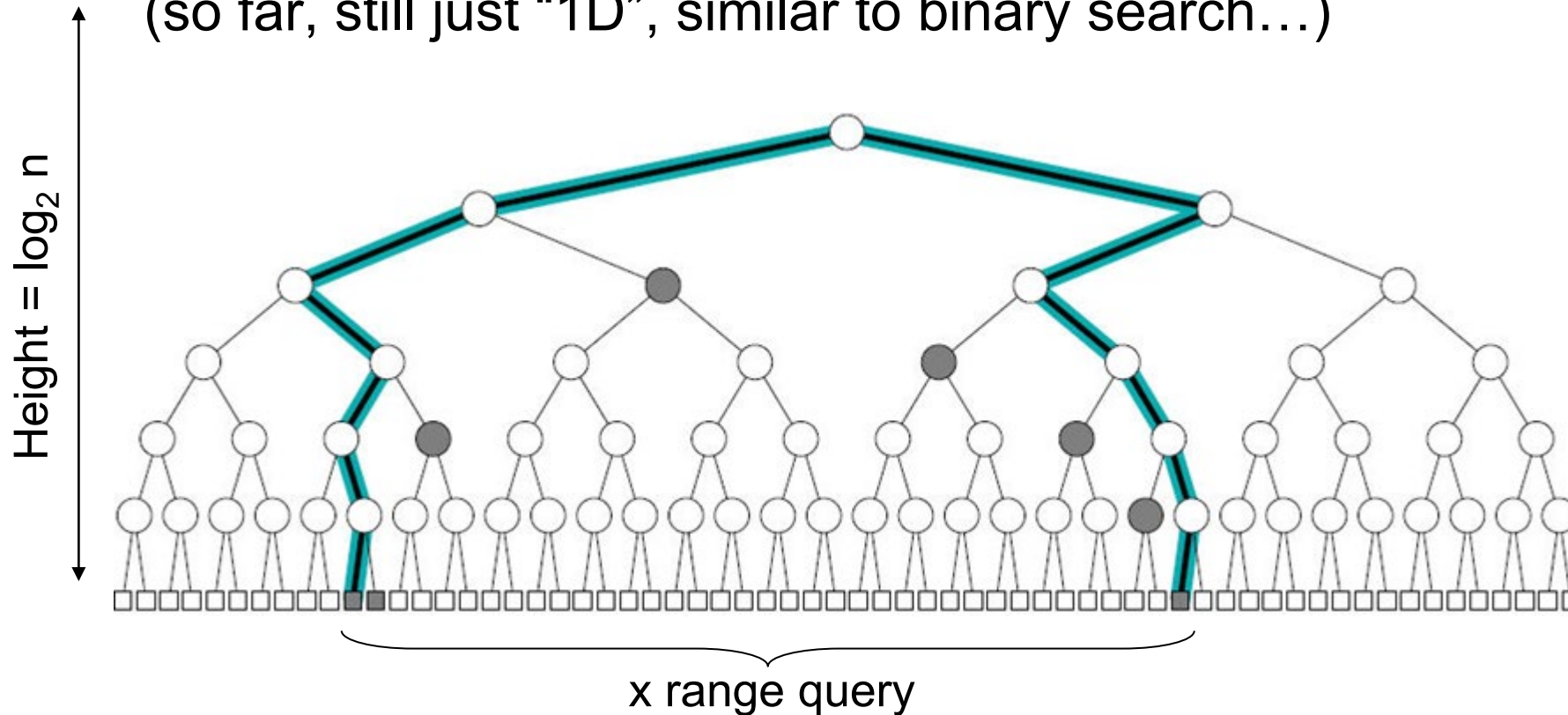
Composing Dimensions: Range Trees

- Sort points by x coordinate and build a segment tree “on top of” this ordering.



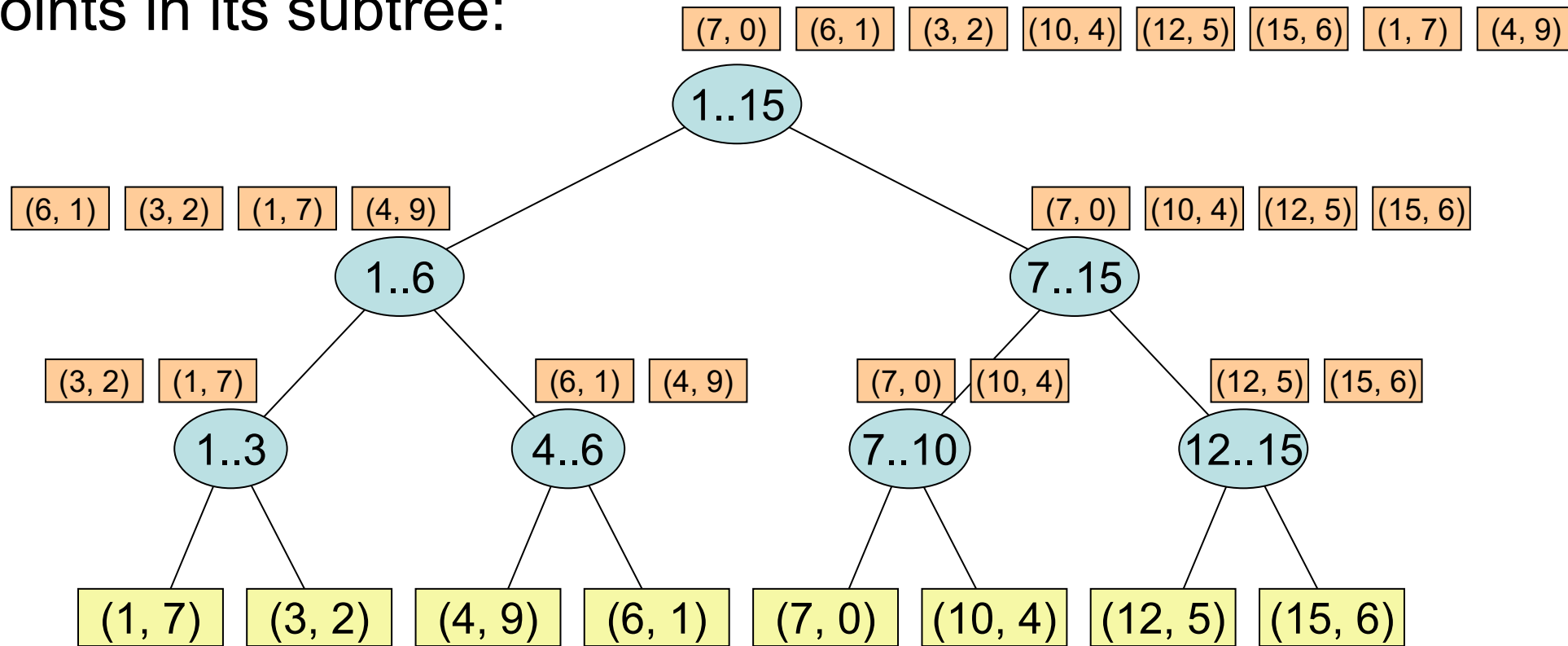
Composing Dimensions: Range Trees

- We can answer a range query in x with a collection of at most $2\log_2 n$ subtrees.
(so far, still just “1D”, similar to binary search...)



Composing Dimensions: Range Trees

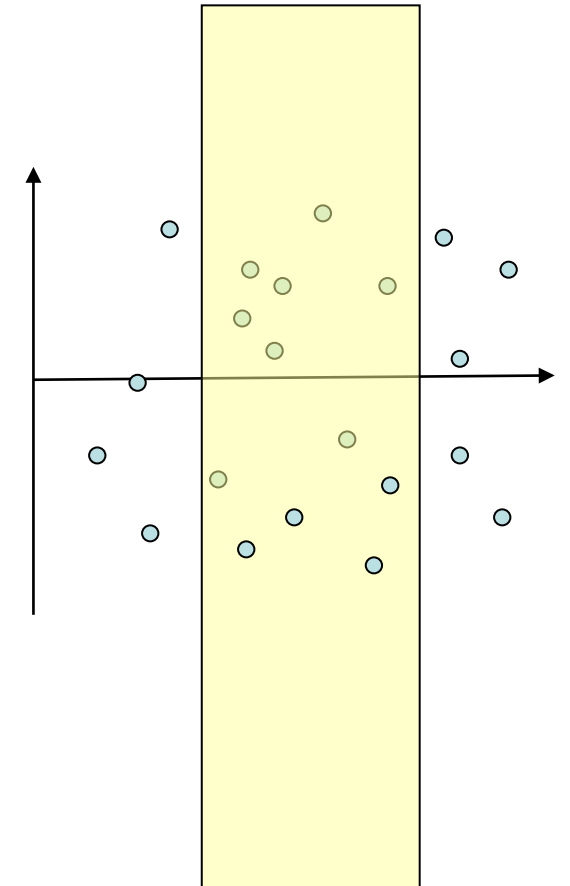
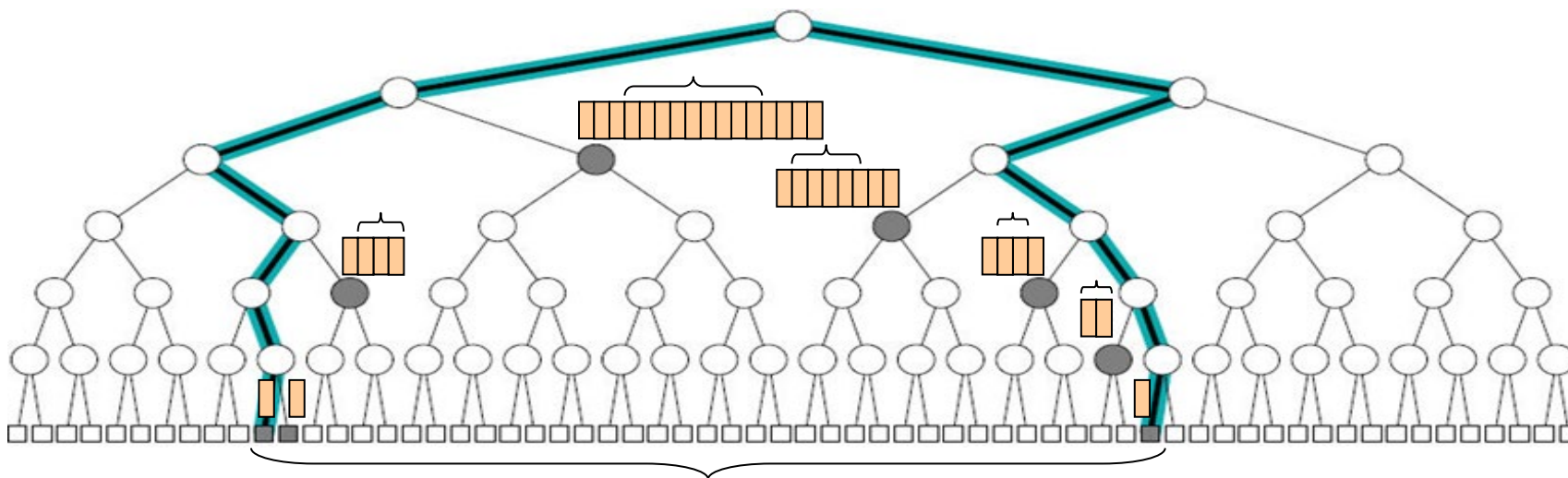
- Augment each node with sorted “y” range tree on points in its subtree:



- Only $O(n \log n)$ space. To build, just run merge sort!

The Range Tree : Query Performance

- 2D range query = $O(\log n)$ 1D queries
- Total query time: $O(k + \log^2 n)$.
- In d dimensions: $O(k + \log^d n)$; $O(n \log^{d-1} n)$ space
 - Can be reduced to $O(k + \log^{d-1} n)$ using a technique called “fractional cascading”



Stabbing Queries: The “Inverse” of Range Queries...

- **Range query:** given data points, tell me points in a query range.
- **Stabbing query:** given ranges, tell me ranges stabbed by a data point.
- Often can be converted back to equivalent range queries...

